How to estimate correlation between metrics from past A/B tests

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The metrics that matter most in experiments are usually the hardest to measure. Let's take the example of *customer transactions* orbookings, a typical metric used in experimentation. If a product team introduces a change in the search page, even if there is a visible impact on upper funnel metrics, such as clicks, this impact will be diluted in later stages of the funnel. These crucial business outcomes, the **goal metrics,** are often sluggish to react. Waiting weeks, or even months, for a conclusive signal slows down product iteration and learning.
That's where **proxy metrics** come in. A well-chosen proxy metric is like a high-sensitivity sensor: it responds quickly to the same forces that ultimately move the goal metric, letting you detect the desired effects sooner.
A reliable proxy must meet two criteria [1]:

1. : the effect of the intervention on the proxy should be easier to observe than on the goal metric. Typically proxies that could be measured close to the intervention are highly sensitive, for example views, clicks or other upper funnel engagement signals.
2. : A positive change in the proxy should consistently predict a positive (or negative) change in the goal. It's not enough to know that clicks and bookings are related — of course they are, you can't book without clicking. What we really care about is whether a product change that *increases clicks* will also *increase bookings*.

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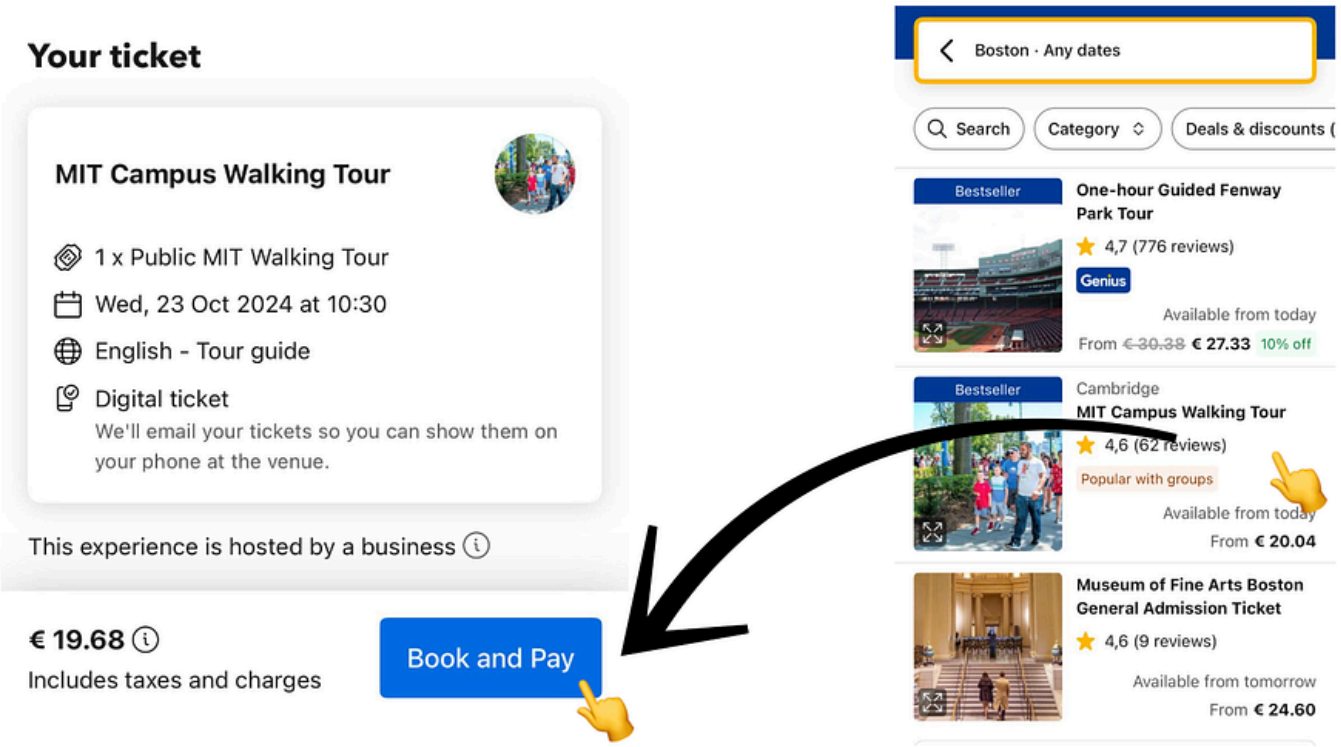

Figure 1: From click to booking — can early signals predict final outcomes?

Naive correlation checks mislead

When trying to quantify how well a proxy predicts a goal, experimenters are often tempted by two intuitive (but flawed!) approaches [2].

1. Correlating observational user-level data

The simplest approach would be to use observational data: calculate the correlation between user-level proxy and goal outcomes, for example clicks and bookings as shown in Table 1.

Table 1: Observational user-level data.

This correlation will be positive: users who book typically click. However, *making a change that will make more people click doesn't necessarily mean they will also book*. Consider clickbait: over-promising eye-catching product cards might boost clicks on irrelevant products while actually reducing bookings by attracting users who realise the content doesn't match their expectations.

2. Correlating observed effects across past experiments

Suppose we have enough (relevant) past experiments tracking both metrics. Figure 2 displays the observed effect on the proxy (e.g., change in clicks) against the observed effect on the goal (e.g., change in bookings) for 100 experiments.

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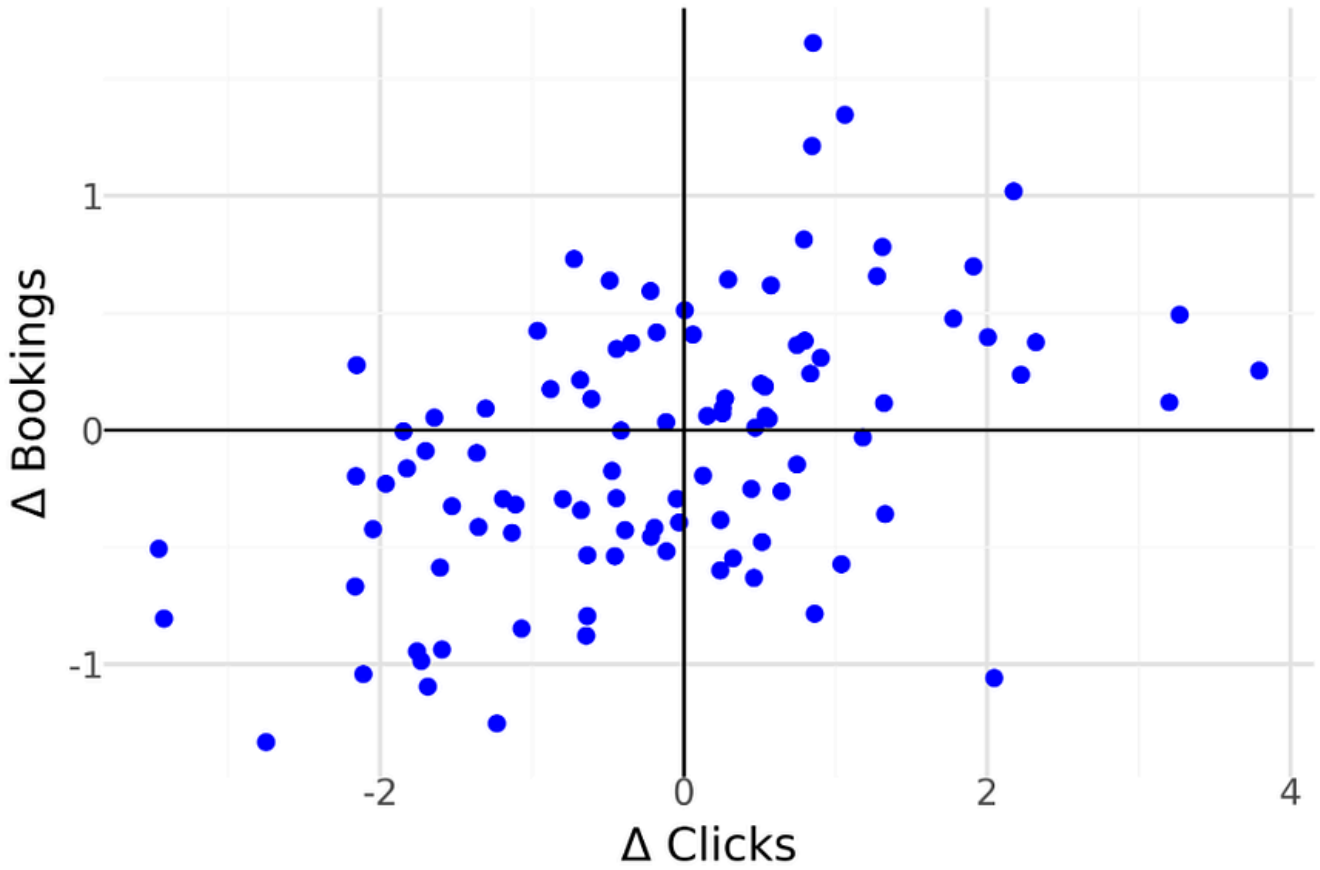

Figure2: Observed effects of clicks (proxy) vs bookings (goal) across experiments — correlation of 0.5 looks promising at first glance.

The scatter plot looks encouraging, with a positive slope and a correlation of 0.5. However, we still can’t conclude that changes improving the proxy will also improve the goal. The problem becomes obvious when we look at A/A tests — experiments with no actual changes.

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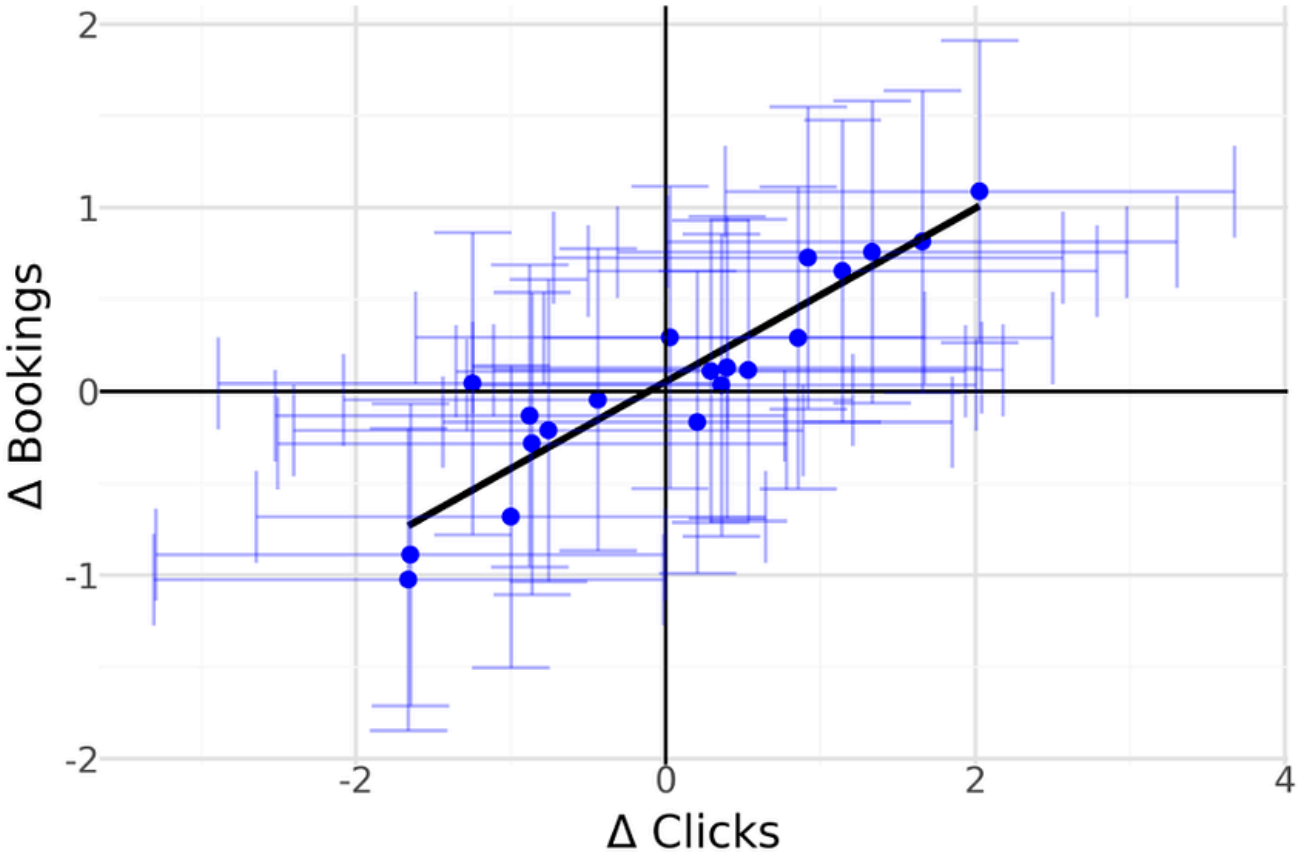

Figure 3: Even in A/A tests (no true effect), correlated noise creates a strong correlation. Plot also displays 90% confidence intervals.

Even with no treatment effects, A/A tests show a strong positive correlation (Figure 3). Why? If metrics are positively correlated on the user-level, then the experiment variant that happens to have more visitors with high proxy will also tend to have more visitors with high goal, even in the absence of effects. This apparent correlation between effects is caused by **correlated sampling/measurement error.**

As a result, observed treatment effects are contaminated by this error, which can sometimes overwhelm the real correlation. In the next section, we show how to correct for this noise and recover more accurate estimates of the true effects and their correlation.

A better way: combine the two naive approaches

To understand why these naive approaches fail — and how to fix them — we need to break down what an observed treatment effect is actually made of. The effect we see is the sum of two distinct components [4]:

1. The real impact of the change.
2. Random variation due to sampling, usually described by standard errors or confidence intervals.

We will use letters P and G to denote proxy and goal metrics.

True effects

For a set of “comparable” past experiments, we assume the true (but unobserved) treatment effects on the proxy and goal follow a bivariate normal distribution:

Equation 1: Bivariate normal distribution of (unobserved) true treatment effects.

In Equation 1, (t_i^P, t_i^G) denote the true treatment effects for proxy and goal metrics in experiment i. (μ^P, μ^G) denote the mean true effects across experiments, typically close to zero. Σ_t is the covariance matrix, whose diagonal contains the variances of the true effects for proxy and goal, and the off-diagonal entry is the covariance we aim to estimate.

Measurement error

With a large enough sample and proper experimental design, the Central Limit Theorem implies that measurement errors approximately follow a bivariate normal distribution:

Equation 2: By the Central Limit Theorem, the noise is approximately bivariate-normal with mean 0.

This noise is mean-zero (unbiased) and has a covariance matrix consisting of standard deviations that are equal to the standard errors of our metrics and correlation equal to the visitor-level correlation between the metrics.

Observed effects

An observed treatment effect is simply the sum of the true effect and its measurement noise. Since both components are bivariate normal, their sum is too.

Equation 3: Distribution of observed effects is again bivariate normal due to being a sum of two bivariate normal distributions, true effects and noise.

This decomposition reveals why naive approaches are biased: the covariance (and correlation) of observed effects is influenced by covariances of the true effects, but also the noise.

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The Figure 4 below shows each of the components for the clicks vs. bookings example.

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Figure 4: True treatment effects (TEs) are slightly negatively correlated (left), measurement errors are strongly positively correlated (middle), resulting in observed effects that appear positively correlated (right). Each dot represents one experiment.

This reveals two interesting scenarios:

1. : When true effects are large relative to noise, the observed correlation approaches the true correlation. The noise becomes negligible.
2. : When true effects are small relative to noise, the observed correlation is dominated by the noise correlation — essentially just reflecting the visitor-level correlation between metrics. This explains why A/A tests show spurious correlations: with zero true effects, we are only observing correlated measurement noise.

Estimating the correlation between true effects

Now that we understand how measurement noise behaves statistically, we can adjust for it. Since the noise and true effects are independent, their covariance matrices are additive. This means we can isolate the true effect covariance by simply **subtracting the noise covariance from the observed covariance.** This method is also known as the **Total Covariance Estimator** [2]:

Equation 4: The covariance of true effects is estimated by subtracting the noise covariance from the observed effects covariance.

This gives us the covariance matrix of true effects, from which we can extract:

- The correlation between true treatment effects.
- The variances of true effects for the proxy and goal metrics.

Here’s what this decomposition looks like in our clicks vs. bookings example:

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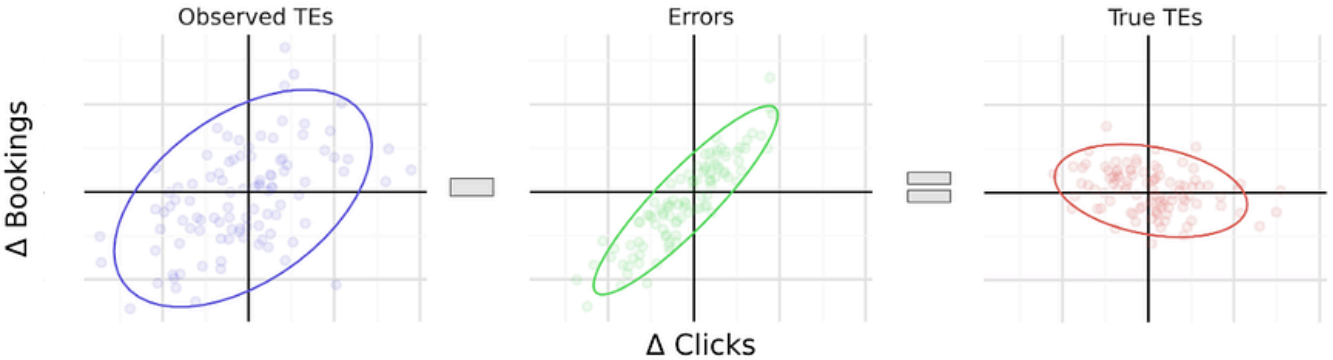

Figure 5. Decomposing observed treatment effects (TEs) into measurement error and true treatment effects (ellipses show 90% confidence regions from covariance matrices). Dots (per-experiment values) are observed only for the observed effects.

Generalised solution

The Total Covariance Estimator works when all experiments share the same covariance (i.e. equal sample sizes, metric variance, and user-level correlations). One way to enforce this is to downsample experiments — but this discards useful information.

A more general approach is to use a **probabilistic multilevel model** that:

- accounts for experiment-specific noise covariance matrices,
- allows for more flexible distributions (e.g., t-distributions for fat tailed effects),
- uses priors to stabilise estimates.

While more complex to implement and fit (requires numerical methods, such as MCMC), this approach provides more accurate estimates and uncertainty quantification. A nice property of this approach is that it simultaneously estimates Σ_t and uses that information to obtain posterior estimates of the effects per experiment, which are more accurate than the unadjusted observed effects [3]. For technical details, see [1] and [3].

Putting it to practice

Table 2 shows how different estimation methods of the true effects yield wildly different results, often leading to contradicting conclusions about which metric to target.

Table 2: Comparison of correlation estimates. The “Method” column lists the statistical approach, “Correlation” shows the resulting value, and “Target” shows the implied primary metric. Note how naive methods produce a misleadingly positive correlation, while the bias-corrected estimate correctly identifies the true negative relationship.

The naive methods suggest a strong positive correlation, implying we should optimize for more clicks. However, the bias-corrected Total Covariance Estimator tells a completely different story: the estimate is much closer to the ground truth, correctly revealing that increasing clicks actually tends to decrease bookings.

To illustrate the damage of targeting clicks assuming the data generating process remains the same, we use the following metrics:

- : The percentage of experiments deemed successful (significantly positive).
- : The cumulative change number of bookings (our goal metric) across all shipped experiments. We normalise this value to provide an intuitive comparison between the different primary metrics.
- (False sign rate): The percentage of shipped experiments that actually decrease bookings.

The results are sobering. While targeting clicks nearly doubles the ship rate, the true impact on bookings is catastrophic — over 60% of these shipped experiments would actually harm the business.

Table 3: Outcomes of choosing bookings vs. clicks as the primary metric. While clicks yield a higher ship rate, they lead to negative cumulative impact on bookings and a much higher risk of shipping harmful experiments.

Our example might look extreme, however similar situations can arise in low SNR environments, which are common in A/B testing.

In this post we've focused on a single primary metric for simplicity, but the same framework extends naturally to weighted combinations of proxy metrics [1, 2].

Metric correlation at Booking.com

At Booking.com, we're standardising how we estimate metric correlations across the company. Our internal infrastructure supports both the TotalCovariance Estimator and the more flexible probabilistic multilevel model. This empowers teams to assess the true correlation between metrics using their own historical experiments and domain expertise, choosing the best approach for their needs.

Of course, correlation estimation is only one part of the broader challenge. The real objective is a decision framework that explicitly balances a metric's sensitivity, business impact, and error rates when selecting proxies and making shipping decisions. Building such a framework — resembling the one illustrated above — is an ongoing effort, but it is critical to ensuring that product decisions are based on signals we can trust.

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